PFD Team Project P9G4
Lee Guan Feng (2603062)
Xavier Heng (2603020)
Bryan Ow (2603026)
Arvin Lum (2605736)
Bryan Lee (2604464)
Jeremy Goh (2605327)

PhishReport

Project Initial Details Submission | 11 September 2026
CLI-based phishing and exposure assessment

PROBLEM STATEMENT AND TARGET USERS

Day to day, many may struggle to recognise phishing messages and decide what to do after clicking a link or sharing information. PhishReport will help users assess suspicious messages through a command-line interface (CLI). It will explain message indicators, combine them with the user's reported actions, and provide a prioritised response checklist. Its purpose is to not only certify that a message is safe, but to also support informed action.

The scope covers pasted email, SMS and chat text. Users can analyse messages, view saved reports, filter by priority, and update report status. The application will not connect to inboxes, visit links, open attachments or automatically change accounts.

Since the user will only be interacting with the AI through a CLI interface instead of a dedicated Guided User Interface (GUI), our target users will mainly focus on students and young adults who do not require much visual assistance to navigate and are somewhat tech-savvy enough to quickly adapt to the CLI.

USER INPUTS

Users enter the message text, channel (email/SMS/chat), optional sender information (email address/phone number), and whether they were sent a link, a file or requested for user information. If information was submitted, users select its category, such as password, one-time code or personal details; they must not enter actual secrets. Blank messages, unsupported choices and inconsistent answers will be rejected with a re-prompt. Development and demonstrations will use fictional or sanitised messages.

USE OF AI

Every new report will pass through the Kimi K3 API. To maintain a zero-cost project, implementation is subject to verified free API access and its usage limits. The AI will interpret

the message's meaning and return structured JSON containing impersonation indicators, credential or payment requests, urgency tactics, supporting excerpts and an insufficient-context flag. This contextual interpretation is essential to the application's assessment.

The AI manager will build the prompt, call the API, and validate required fields, types and allowed values. Message text will be treated as untrusted data, not instructions. Invalid responses will receive a bounded retry; API failures will be logged and marked for retry without crashing or producing a successful assessment. The logic manager will assign the final priority and select a predefined checklist from the validated AI fields and user actions.

BUSINESS RULES

Rules will be evaluated in priority order:

1. **High:** AI identifies a credential request and the user reports submitting a password or one-time code. Recommend account-recovery action through official channels and reporting to IT.
2. **Medium:** AI identifies suspicious indicators and the user clicked or downloaded something, without triggering the high-priority rule. Recommend stopping further interaction and seeking IT guidance.
3. **Review:** Suspicious indicators are present and no higher-priority rule applies. Recommend independent verification and reporting.
4. **Insufficient information:** AI cannot assess the context and no higher-priority rule applies. Request clarification.
5. **No clear indicators:** None of the above applies. Explain that this is not a guarantee of safety.

The procedural Python application will use four managers: I/O for terminal interaction and all printing; AI for API processing; logic for decisions; and data for JSON persistence and queries. Records will load on startup, with missing or corrupt files handled gracefully. Saved assessments will retain their validated AI output. Docker, automated tests, feature branches and reviewed pull requests will support reproducible development.

TEAM REPOSITORY DETAILS

GitHub repository URL: https://github.com/lgf2111/phish-report
All team members and lab-in-charges will be added as collaborators. Team members will clone the shared repository and commit the initial details document.